Chapter 1

Numerical sequences

Competences: search, model, represent, calculate, communicate

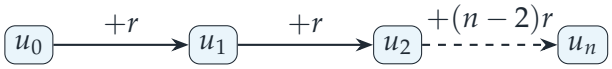
1.1 Reference course

A numerical sequence combines each natural integer n a number u_n. The index n means a rank; it does not must not be confused with the value u_n.

1.1.1 Arithmetic Sequence

Key ideas

A sequence (u_n) is **arithmetic** where there is a real r such as, for all n, $u_{n+1} = u_n + r$. The number r is the common difference / ratio. If the first term is u_0, then $u_n = u_0 + nr$. If the first term is u_1, then $u_n = u_1 + (n-1)r$.



Three numbers x, y, z, written in this order, are three consecutive terms of an arithmetic sequence if $y - x = z - y$, i.e. if $y = \dfrac{x+z}{2}$. This last quantity is the arithmetic mean of x and z.

Method

To model an additive situation: 1) define u_n and specify the unit; 2) identify the first term; 3) translate constant increase or decrease by common difference / ratio r ; 4) Write the term 5) answer with a contextualized sentence.

Worked example

A business sells 480 units the first month and then 25 more units each month. $u_1 = 480$ and $r = 25$, we get $u_n = 480 + 25(n-1)$. Thus $u_{12} = 755$: trade provides 755 units sold in the twelfth month.

1.1.2 Sum of arithmetic terms

For $n+1$ terms of u_0 to u_n,

$$u_0 + u_1 + \cdots + u_n = (n+1)\frac{u_0 + u_n}{2}.$$

The formula is stored in the form of: number of terms multiplied by the average of the first and last term .

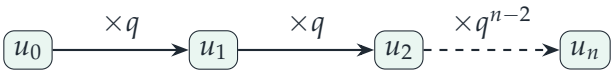
Frequent error

In $u_3 + u_4 + \cdots + u_{12}$, there are $12 - 3 + 1 = 10$ terms, not 12.

1.1.3. Geometric Sequence

Key ideas

A sequence (u_n) is **geometrical** where there is a real q such that $u_{n+1} = qu_n$. If the first term is u_0, then $u_n = u_0 q^n$. If the first term is u_1, then $u_n = u_1 q^{n-1}$.



An increase of t % corresponds to $q = 1 + \dfrac{t}{100}$; a decrease of t % à $q = 1 - \dfrac{t}{100}$. Three positive terms x, y, z are consecutive in a geometric sequence if $y^2 = xz$; $y = \sqrt{xz}$ is their geometric mean.

For $q \neq 1$,

$$u_0 + u_1 + \cdots + u_n = u_0 \frac{1 - q^{n+1}}{1 - q}.$$

1.1.4 Algorithmic: calculate a sum

```
def sum_squares(n):
    s = 0
    for k in rows (1, n + 1):
        s = s + k**2
    return s
```

Accumulator `s` contains the partial sum; the counter `k` travels the entires from 1 to n.

Session 1 — Arithmetic sequence: recognize and model

Numerical sequences

15/09/2026 • B : 2 h – C : 1 h

Objectives of the session

· Recognize an additive evolution.

· Determine the common difference / ratio.

· Define a sequence in a context.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier coefficient of a 10 per cent increase . . . . . .

2. Calculate u_4 if $u_0 = 14$ and $u_{n+1} = u_n + 3$. . . . . .

3. How many terms of u_3 to u_{15} ? . . . . . .

4. Give the coefficient of a decrease of 8 per cent . . . . . .

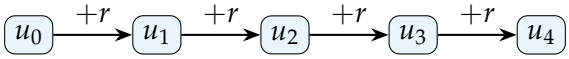
5. Give common difference / ratio for $12, 12 + 4, 12 + 2 \times 4, \ldots$ . . . . . .

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: recognize an additive evolution and determine the Right.

Research activity

An association has 320 members and plans to win 18 each year. u_n the number of staff after n years. u_0, u_1, u_2. What operation is moving from one term to the next? Write a relationship between u_{n+1} and u_n, then plan u_5.



Trace written

A sequence is arithmetic when $u_{n+1} - u_n = r$ is constant. In a situation, an increase absolute constant leads to an arithmetic sequence.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Next steps (u_n) checks $u_0 = 110$ and $u_{n+1} = u_n + 3$. Write u_n then calculate u_{12}.

2. A quantity is worth 420 and a decrease of 3 per cent per period. v_n and calculate v_5.

3. For each sequence, specify whether the model is arithmetic or geometric and justify by a difference or quotient.

Reference in the manual

Supplements: Calao, courses p. 18·21; exercises p. 24, n° 1·3.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 24, exercise 5

For each of the sequences, specify whether it is arithmetic and, if so, indicate its common difference.

a. For the integer n : $u_n = 2(n-5) + 3n$.

b. For the integer n : $v_n = n^2 + n - 1$.

c. For the integer n : $w_n = -4 + 3n$.

Personal work

Resuming Exercise 2 with the following data:
125 and rate or common difference / ratio by 3. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Session 2 — Arithmetic sequence: general term

Numerical sequences

17/09/2026 • B : 1 h – C : 2 h

Objectives of the session

· Moving from recurrence to general term.

· Calculate a remote term.

· Interpret the row.

Fluency · 6 to 8 minutes, without calculator

1. Reduce 3 $x + 5 - x + 2$. .....

2. How many terms of u_3 to u_{15} ? .....

3. Give the coefficient of a decrease of 8 per cent .....

4. Give common difference / ratio for $12, 12 + 5, 12 + 2 \times 5, \ldots$ .....

5. Calculate u_4 if $u_0 = 15$ and $u_{n+1} = u_n + 3$. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: go from recurrence to general term and calculate a distant term.

Research activity

A store has 1,250 active loyalty cards at range 0 and earns 75 per month. all terms, determine strength at rank n, then rank 18. Explain the role of 18 in calculation.

Trace written

If (u_n) is arithmetical of common difference r, then $u_n = u_0 + nr$ or $u_n = u_1 + (n-1)r$, depending on the rank of First term.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Next steps (u_n) checks $u_0 = 120$ and $u_{n+1} = u_n + 4$. Write u_n then calculate u_{12}.

2. A quantity is worth 440 and a 4% decrease per period. v_n and calculate v_5.

3. For each sequence, specify whether the model is arithmetic or geometric and justify by a difference or quotient.

Reference in the manual

Supplements: Calao, courses p. 18·21; exercises p. 24, n° 5·8.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 24, exercise 6

Let (l_n) the arithmetic sequence of common difference 2 and first term: $l_0 = -1$.

a. Express l_n for all natural n.

b. Calculate l_{35} and l_{64}.

Personal work

Resuming Exercise 3 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Session 3 — Sum of terms of an arithmetic sequence

Numerical sequences

22/09/2026 • B : 2 h – C : 1 h

Objectives of the session

· Count the terms of an amount.

· Apply the formula.

· Control the order of magnitude.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 20 per cent of 120 .....

2. Give the coefficient of a decrease of 8 per cent .....

3. Give common difference / ratio for $12, 12+6, 12+2\times 6, \ldots$ .....

4. Calculate u_4 if $u_0 = 16$ and $u_{n+1} = u_n + 3$. .....

5. How many terms of u_3 to u_{15} ? .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: count the terms of a sum and apply the formula.

Research activity

A company pays a monthly premium: 45 euros the first month, then 5 euros each calculate the total paid over the first 12 months in three ways: explicit sum, First-last grouping, formula.

Trace written

A sum of arithmetic terms is a $q - p + 1$ terms.

Number of terms $\times \dfrac{\text{First} \quad + \text{last}}{2}$. From u_p to u_q, there is

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Next steps (u_n) checks $u_0 = 130$ and $u_{n+1} = u_n + 5$. Write u_n then calculate u_{12}.

2. A quantity is worth 460 and a 5% decrease per period. v_n and calculate v_5.

3. For each sequence, specify whether the model is arithmetic or geometric and justify by a difference or quotient.

Reference in the manual

Supplements: Calao, courses p. 20·21; exercises p. 24, n° 11-13, then p. 28, n° 43·44. reproduced in this session is identified below by its page and number; the reference allows also find the figures and digital resources of the manual.

Manuel Calao · p. 24, exercise 9

a. Calculate the sum of the first 500 non-zero integers $\sum_{i=1}^{500} i$.

b. Calculate the sum of the totals from 35 to 150: $\sum_{k=35}^{150} k$.

Personal work

Resuming Exercise 2 with the following data:
135 and the rate or common difference / ratio by 5. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Session 4 — Geometric Sequence: recognize and model

Numerical sequences

24/09/2026 • B : 1 h – C : 2 h

Objectives of the session

· Recognize multiplicative evolution.

· Switch from a rate to a coefficient.

· Compare arithmetic and geometric.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier coefficient of a 25 per cent increase . . . . . .

2. Give common difference / ratio for $12, 12+7, 12+2\times 7, \ldots$ . . . . . .

3. Calculate u_4 if $u_0 = 17$ and $u_{n+1} = u_n + 3$. . . . . .

4. How many terms of u_3 to u_{15} ? . . . . . .

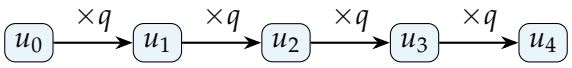
5. Give the coefficient of a decrease of 8 per cent . . . . . .

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: recognize a multiplicative evolution and pass a rate at a coefficient.

Research activity

An online shop increases its number of visitors by 6% per month from 8,000 visitors. Calculate the first three values. Compare differences then quotients. Identify nature I'll do it later.



Trace written

A sequence is geometric when $\dfrac{u_{n+1}}{u_n} = q$ is constant (non-zero terms).

constant leads to a geometric sequence.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Next steps (u_n) checks $u_0 = 140$ and $u_{n+1} = u_n + 6$. Write u_n then calculate u_{12}.

2. One quantity is worth 480 and decreases by 6 per cent per period. v_n and calculate v_5.

3. For each sequence, specify whether the model is arithmetic or geometric and justify by a

difference or quotient.

Reference in the manual

Supplements: Calao, courses p. 23·25; exercises p. 25, n° 14·18.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 25, exercise 19

Let (l_n) the geometric sequence of common ratio 2 and first term: $l_0 = -1$.

a. Express l_n for all natural n.

b. Calculate l_{12} and l_{20}.

Personal work

Resuming Exercise 3 with the following data:
140 and the rate or common difference / ratio by 6. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Session 5 — Geometrical sequence: general term and sum

Numerical sequences

29/09/2026 • B : 2 h – C : 1 h

Objectives of the session

· Write the general term.

· Calculate a geometric sum.

· Interpret a cumulative evolution.

Fluency · 6 to 8 minutes, without calculator

1. Reduce 3 $x + 8 - x + 2$. .....
2. Calculate u_4 if $u_0 = 18$ and $u_{n+1} = u_n + 3$. .....
3. How many terms of u_3 to u_{15} ? .....
4. Give the coefficient of a decrease of 8 per cent .....
5. Give common difference / ratio for $12, 12 + 8, 12 + 2 \times 8, \ldots$ .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: write the general term and calculate a sum geometrical.

Research activity

A capital of 250 000 Ar is invested at 4% per year, compound interest. C_n after n years. C_6, then the sum $C_0 + \cdots + C_6$ and explain what the latter sum is mathematically represents and why it is not the final capital.

Trace written

If (u_n) is geometrical of common ratio q, $u_n = u_0 q^n$ or $u_n = u_1 q^{n-1}$. For $q \neq 1$, $u_0 + \cdots + u_n = u_0 \dfrac{1 - q^{n+1}}{1 - q}$.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Next steps (u_n) checks $u_0 = 150$ and $u_{n+1} = u_n + 2$. Write u_n then calculate u_{12}.
2. A quantity is 500 and a decrease of 2 % per period. v_n and calculate v_5.
3. For each sequence, specify whether the model is arithmetic or geometric and justify by a difference or quotient.

Reference in the manual

Supplements: Calao, courses p. 25·28; exercises p. 25, n° 19-25, and p. 28, n° 42·44. reproduced in this session is identified below by its page and number; the reference allows also find the figures and digital resources of the manual.

Manuel Calao · p. 25, exercise 22

What's next? (u_n) geometrical common ratio 3 and first term: $u_0 = 5$.

a. Express u_n for all natural n. Deduct from it u_7.

b. Calculate: $S = \sum_{i=0}^{7} u_i = u_0 + u_1 + \cdots + u_7$.

Personal work

Resuming Exercise 2 with the following data:
145 and the rate or common difference / ratio by 7. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Session 6 — Assessment and formative evaluation

Numerical sequences

01/10/2026 • B : 1 h – C : 2 h

Objectives of the session

· Choose the right model.

· Mobilize formulae.

· Analyze errors.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5 per cent of 180 .....
2. How many terms of u_3 to u_{15} ? .....
3. Give the coefficient of a decrease of 8 per cent .....
4. Give common difference / ratio for $12, 12+9, 12+2\times 9, \ldots$ .....
5. Calculate u_4 if $u_0 = 19$ and $u_{n+1} = u_n + 3$. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today.
Before the activity, reformulate the keywords: choose the right model and mobilize formulas.

Research activity

Classify eight situations according to three categories: arithmetic, geometric, neither one nor the other.
each situation, highlight the word or data that justifies the choice, then choose the appropriate formula.

Trace written

Before calculating, identify: type of sequence, first row, common difference / ratio, term or amount requested.
conclusion specifies unity and rank.

Worked example

Retrieve activity data and write the solution in order: definition of objects,
property used, exact calculation or approach, then response phrase. The model response figure
in the teaching edition.

Training · essential

1. Next steps (u_n) checks $u_0 = 160$ and $u_{n+1} = u_n + 3$. Write u_n then calculate u_{12}.
2. A quantity is 520 and a decrease of 3% per period. v_n and calculate v_5.
3. For each sequence, specify whether the model is arithmetic or geometric and justify by a difference or quotient.

Reference in the manual

Supplements: Calao, p. 29, No 49-52. The exercise reproduced in this session is identified below. by its page and number; the reference also makes it possible to find the figures and the digital resources of the manual.

Manuel Calao · p. 29, exercise 49

For each proposal, state whether it is true or false. Justify the answer.

1. A capital of 5,000 € is placed on a simple interest account at the annual interest rate of This situation can be modelled by a geometric sequence of common ratio 1,04.
2. The number of sparrows declines by 20% per year in one region. reintroduction of 200 sparrows each year. This situation can be modelled by a follow-up arithmetical common difference 200.
3. The mass of household waste falls by 10% per year. This situation can be modelled by a geometric sequence of common ratio 1.1.
4. The production of a company increases weekly by 200 items produced. This situation can be modelled by an arithmetic sequence of common difference 200.
5. The number of subscribers to a youtuber increases by 15% every month. be modelled by an arithmetic sequence of common difference 15.
6. The annual rent of an apartment increases by 320 € Every year, this situation can be be modelled by an arithmetic sequence of common difference 320.
7. On 1 1st January of a year, Enzo places 1 000 € with compound interest on an account the rate of which annual is equal to 2 % and 1 1st January of each following year, it places 200 € on this account. This situation can be modelled by a geometric sequence of common ratio 1,02.

Personal work

Resuming Exercise 3 with the following data:
150 and the rate or common difference / ratio by 2. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.